Haoyu Zhai

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RESEARCH INTERESTS

Trustworthy Machine Learning; LLM Agent Security; Reliable AI Systems; Out-of-Distribution and Novelty Detection; Clustering and Anomaly Detection.

EDUCATION

Shanghai Normal University 2022.09 - 2025.06 | Shanghai, China

Master of Engineering in Computer Science and Technology
Major courses: Digital Image Processing (90), Digital Signal Processing (90), Computer Network and Simulation (94)

Nanjing University of Aeronautics and Astronautics Jincheng College 2017.09 - 2021.06 | Jiangsu, China

Bachelor of Engineering in Transportation
Major courses: Linear Algebra (93), Advanced Mathematics (98), Probability Theory and Mathematical Statistics (100)

PUBLICATIONS

[1] **Haoyu Zhai**, Jie Yang, Haotao Guo, Bin Wang, Yan Ma. “Density-increment and cut-edge optimized clustering via minimum spanning forest.” **Neurocomputing**, 2026. DOI: 10.1016/j.neucom.2026.132957

[2] Zexuan Fei, **Haoyu Zhai**, Jie Yang, Bin Wang, Yan Ma. “Discovering generalized clusters with adaptive mixture density-based clustering.” **Knowledge-Based Systems**, 2025. (Co-first authors). DOI: 10.1016/j.knosys.2025.113250

[3] **Haoyu Zhai**, Zexuan Fei, Yan Ma. “A Novel Outlier Detection Algorithm Based on Symmetry and Distance Ratio.” **International Conference on Pattern Recognition**, 2024. DOI: 10.1007/978-3-031-78192-6_22

RESEARCH EXPERIENCE

MAGA: LLM Agent System for Malware and Security Analysis 2025.07 - Present | Qihoo 360

Role: Main developer / AI Algorithm Engineer
- Built an LLM-based autonomous agent system for malware and security triage, using ReAct-style planning, tool orchestration, and Docker-based sandbox isolation.
- Integrated binary/PE inspection, Office/PDF malware triage, file operations, shell/Python/Node.js execution, and structured evidence collection into a unified agent workflow.
- Developed batch multi-agent dispatch and a FastAPI WebUI for repeatable experiments; the system serves as a testbed for reliable tool-chain execution and runtime failure detection.

NSFC Project: Graph-Based Clustering and Image Recognition 2022.09 - 2025.06 | Shanghai Normal University

Role: Algorithm designer / core contributor
- Designed and evaluated graph-based clustering, density-based clustering, and outlier detection algorithms for large-scale image and tabular datasets.
- Proposed topology-aware splitting/merging strategies and symmetry-distance outlier scoring; contributed to 3 closely related peer-reviewed publications.
- Contributed to related patent/application materials from the project under the supervision of senior collaborators.

AWARDS AND HONORS

Outstanding Graduate Award, Shanghai Normal University, 2025.
Shanghai Normal University Scholarship, 2022, 2023, 2024.
NUAA Jincheng College Scholarship, 2017, 2018, 2019, 2020.

SKILLS

- **Software and Coding Skills:** Python, C/C++, Matlab, Git, Linux, Docker, FastAPI, LaTeX, Typst.
- **Machine Learning:** PyTorch, OpenCV, Scikit-learn, clustering, anomaly detection, OOD/novelty detection, computer vision.
- **LLM and Agent Systems:** LLM agents, ReAct-style tool orchestration, Docker sandboxing, MCP, RAG pipelines, vector databases, Milvus.